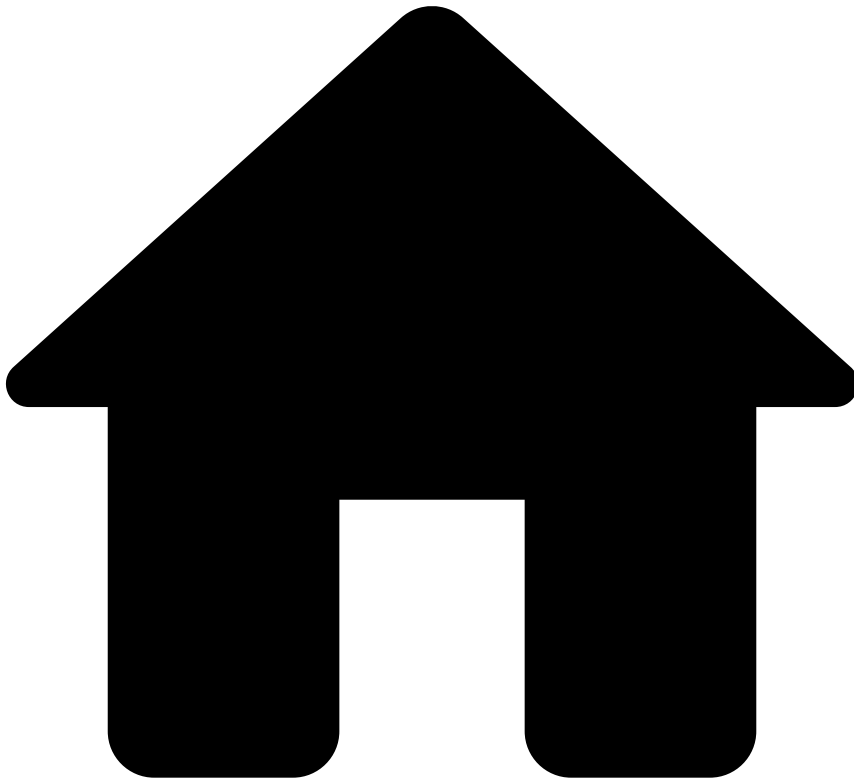


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- [References](#)
- [Rules Projects](#)
- Digital Activity - Fraud Rules

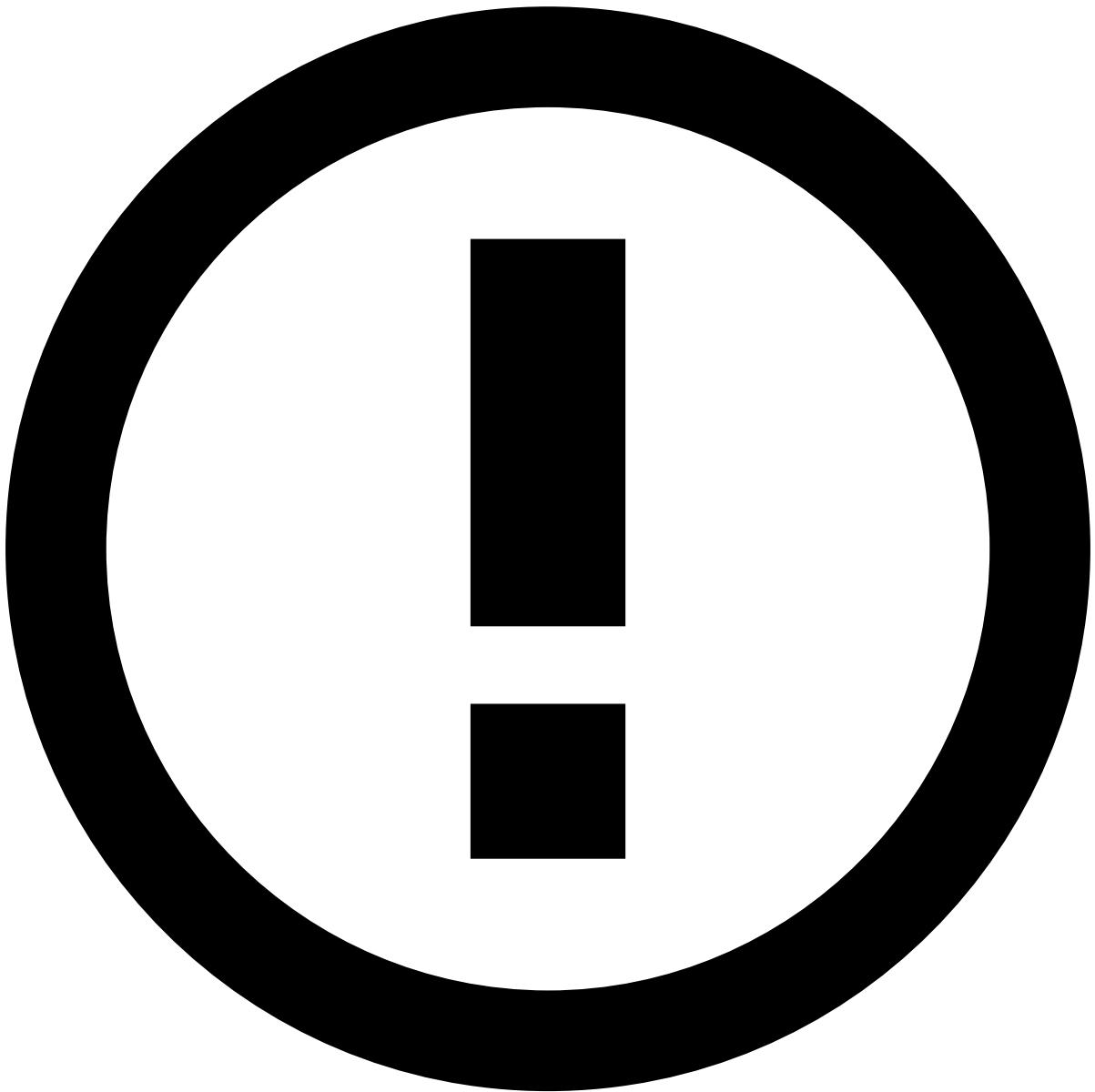
On this page

Digital Activity - Fraud Rules

Was this page helpful? 

Description

Fraud Rules look for common fraud patterns in digital events (non-financial events). If triggered, the rules included in this rules project mark the event as `high` , `medium` or `low` risk. Decision Rules block will then provide the final recommendation to whether a financial institution should accept, challenge, or decline a given event, according to the correspondent risk level. Check [Decision Rules](#) for more details about Feedzai's final recommendation.



info

Note that events marked as `medium` or `low` risk do not trigger an alert in TFB's pre-packaged resource, but this option is configurable.

Rules Project Metadata

- *Target Variable:* risk_level
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Digital Activity - FR1

Comment

Login attempt from a Device Id linked to multiple customers (last 7d) and with a high number of logins (last 90m)

Actions

- [DA-FR1]

Variables

```
max_customers_per_device =  
digital_activity__rules_thresholds['max_customers_per_device'].threshold;  
  
max_logins_per_device = digital_activity__rules_thresholds['max_logins_per_device'].threshold;
```

Condition

```
//Filters  
(event.device_id ?? '' != '') and (event.customer_id ?? '' != '')  
and  
(event.event_type == 'authentication' and event.challenge_requested != true)  
and  
//Limits  
event.distinct_customer_id_per_device >= max_customers_per_device  
and  
event.count_logins_per_device_id_90m >= max_logins_per_device
```

Explanation

[DA-FR1] - Medium Risk - Multiple customers per device, last 7d,
({{event.distinct_customer_id_per_device}}) above [{localvar.max_customers_per_device}] and logins, last
90m, ({{event.count_logins_per_device_id_90m}}) above [{localvar.max_logins_per_device}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR2

Comment

Login attempt from an IP Address linked to multiple customers and with a high number of login attempts, in the last hour

Actions

- [DA-FR2]

Variables

```
max_customers_per_ip_addr =  
digital_activity__rules_thresholds['max_customers_per_ip_addr'].threshold;
```

```
max_logins_per_ip_addr =
digital_activity__rules_thresholds['max_logins_per_ip_addr'].threshold;
```

Condition

```
//Filters
(event.device_ip_address ?? '' != '') and (event.customer_id ?? '' != '')
and
(event.event_type == 'authentication' and event.challenge_requested != true)
and
//Limits
event.distinct_customer_id_per_devsr >= max_customers_per_ip_addr
and
event.count_logins_per_device_ip_asr >= max_logins_per_ip_addr
```

Explanation

[DA-FR2] - Medium Risk - Multiple customers per IP, last 1h, ({event.distinct_customer_id_per_devsr}) above [{localvar.max_customers_per_ip_addr}] and logins attempts ({event.count_logins_per_device_ip_asr}) above [{localvar.max_logins_per_ip_addr}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR3

Comment

Login attempt from an unseen device with a high number of failed login attempts, in the last 90m

Actions

- [DA-FR3]

Variables

```
max_failed_logins_per_device =
digital_activity__rules_thresholds['max_failed_logins_per_device'].threshold;

min_succ_logins_per_account =
digital_activity__rules_thresholds['min_succ_logins_per_account'].threshold; // min of 10 logins
per week
```

Condition

```
//Filters
(event.device_id ?? '' != '')
and
(event.event_type == 'authentication' and event.challenge_requested != true)
and
//Account Behavior
event.count_succ_logins_per_accounsr >= min_succ_logins_per_account
and
event.count_logins_per_device_id_1m == 0
and
//Limits
event.count_failed_logins_per_device >= max_failed_logins_per_device
```

Explanation

[DA-FR3] - Medium Risk - Device Id ({event.device_id}) not seen in the last month and number of failed logins per device, in the last 90m ({event.count_failed_logins_per_device}), is above limit [{localvar.max_failed_logins_per_device}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR4

Comment

Login attempt from an unseen IP Address with a high number of failed login attempts, in the last 90min

Actions

- [DA-FR4]

Variables

```
max_failed_logins_per_ip_addr =
digital_activity__rules_thresholds['max_failed_logins_per_ip_addr'].threshold;

min_succ_logins_per_account =
digital_activity__rules_thresholds['min_succ_logins_per_account'].threshold; // min of 5 logins
per week
```

Condition

```
//Filters
(event.device_ip_address ?? '' != '')
and
(event.event_type == 'authentication' and event.challenge_requested != true)
and
//Limits
event.count_succ_logins_per_accountsr >= min_succ_logins_per_account
and
event.count_logins_per_device_ip_add == 0
and
event.count_failed_logins_per_devisr >= max_failed_logins_per_ip_addr
```

Explanation

[DA-FR4] - Medium Risk - IP Address ({event.device_ip_address}) not seen in the last month and number of failed logins per IP Address, in the last 90m ({event.count_failed_logins_per_devisr}), is above limit [{localvar.max_failed_logins_per_ip_addr}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR5

Comment

High number of failed login attempts per account, in the last minute

Actions

- [DA-FR5]

Variables

```
max_failed_logins_per_account_1min =  
digital_activity__rules_thresholds['max_failed_logins_per_account_1min'].threshold;
```

Condition

```
//Filters  
(event.account_id ?? '' != '')  
and  
(event.event_type == 'authentication' and event.challenge_requested != true)  
and  
//Limits  
event.count_failed_logins_per_accoun >= max_failed_logins_per_account_1min
```

Explanation

[DA-FR5] - Medium Risk - Number of failed logins per account ({event.account_id}), in the last minute ({event.count_failed_logins_per_accosr}), is above limit [{localvar.max_failed_logins_per_account_1min}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR6

Comment

Login attempts from different locations than the account location are above a set threshold, in the last hour

Actions

- [DA-FR6]

Variables

```
max_distinct_locations_per_account =  
digital_activity__rules_thresholds['max_distinct_locations_per_account'].threshold;  
  
customer_region_to_lower_case =  
com.feedzai.commons.udfs.StringTools.lowerCase(event.customer_region);  
device_ip_region_to_lower_case =  
com.feedzai.commons.udfs.StringTools.lowerCase(event.device_ip_region);
```

Condition

```
//Filters  
(event.account_id ?? '' != '') and (event.customer_region ?? '' != '') and  
(event.device_ip_region ?? '' != '')  
and  
(event.event_type == 'authentication' and event.challenge_requested != true)  
and  
customer_region_to_lower_case != device_ip_region_to_lower_case
```

```
and
//Limits
event.distinct_login_locations_per_a >= max_distinct_locations_per_account
```

Explanation

[DA-FR6] - Medium Risk - Login attempts ({event.distinct_login_locations_per_a}) from distinct locations than customer location ({event.customer_region}) are above [{localvar.max_distinct_locations_per_account}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR7

Comment

Multiple account changes in one session

Actions

- [DA-FR7]

Variables

```
max_account_changes = digital_activity__rules_thresholds['max_account_changes'].threshold;
```

Condition

```
//Filters
(event.account_id ?? '' != '') and (event.session_id ?? '' != '')
and
event.event_type != 'authentication'
and
event.challenge_requested != true
and
//Limits
event.count_account_changes_per_sess >= max_account_changes
```

Explanation

[DA-FR7] - Medium Risk - Number of account changes ({event.count_account_changes_per_sess}) per session ({event.session_id}) is above limit [{localvar.max_account_changes}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR8

Comment

Number of new payee requests per account is above a set threshold, in the last minute

Actions

- [DA-FR8]

Variables

```
max_new_payee_per_account_1min =  
digital_activity_rules_thresholds['max_new_payee_per_account_1min'].threshold;
```

Condition

```
//Filters  
(event.account_id ?? '' != '')  
and  
event.event_type == 'new_payee'  
and  
event.challenge_requested != true  
and  
//Limits  
event.count_new_payee_per_account_1m >= max_new_payee_per_account_1min
```

Explanation

[DA-FR8] - Medium Risk - Number of new payee requests per account ({event.account_id}), in the last minute ({event.count_new_payee_per_account_1m}), is above limit [{localvar.max_new_payee_per_account_1min}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR9

Comment

New trusted device added followed by an account change request or new payee request, in the last 24h

Actions

- [DA-FR9]

Variables

N/A

Condition

```
//Filters  
(event.account_id ?? '' != '')  
and  
(event.event_type == 'profile_update' or event.event_type == 'new_payee')  
and  
event.challenge_requested != true  
and  
//Previous event  
event.account_previous_change_is_new == 1
```

Explanation

[DA-FR9] - Medium Risk - New trusted device added followed by account change request ({event.event_type})

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR10

Comment

Failed login attempts per account followed by a successful login and an account change request, in the last 24h

Actions

- [DA-FR10]

Variables

```
max_failed_logins_per_account_24h =  
digital_activity__rules_thresholds['max_failed_logins_per_account_24h'].threshold;
```

Condition

```
//Filters  
(event.account_id ?? '' != '')  
and  
event.event_type != 'authentication'  
and  
event.challenge_requested != true  
and  
//Previous Event  
event.account_previous_activity_is_s == 1  
and  
//Limits  
event.count_failed_logins_per_accosr >= max_failed_logins_per_account_24h
```

Explanation

[DA-FR10] - Medium Risk - Failed login attempts per account ({event.count_failed_logins_per_accosr}) followed by a successful login and an account change ({event.event_type}; {event.event_sub_type}), in the last 24 hours

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR11

Comment

First successful login in the last month, followed by credentials or profile update request, in the last 24h

Actions

- [DA-FR11]

Variables

N/A

Condition

```
//Filters  
((event.account_id ?? '' != '') and event.account_age ?? 0 > 30)  
and  
(event.event_type == 'profile_update' or event.event_type == 'credentials_update')
```

```
and
//Previous event
event.account_previous_activity_is_s == 1
and
//Account historical behavior
event.count_succ_logins_per_accountsr == 0
```

Explanation

[DA-FR11] - Medium Risk - First successful login in 1 month followed by an account update
({event.event_type}; {event.event_sub_type}), in the last hour

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR12

Comment

Login followed by a password reset at unusual hours, in the last 24h

Actions

- [DA-FR12]

Variables

```
max_account_activity_time_of_day_3m_th =
digital_activity__rules_thresholds['max_account_activity_time_of_day_3m_th'].threshold; // max of
5% logins at this time of day

min_account_activity_3m_th =
digital_activity__rules_thresholds['min_account_activity_3m_th'].threshold; // min 10 logins per
week = 120 logins in 3 months

percentage_account_activity_time_of_day_3m =
Maths.safeDivision(event.count_succ_logins_per_account_, event.count_succ_logins_per_accounttr);
```

Condition

```
//Filters
(event.account_id ?? '' != '') and (event.account_age ?? 0 > 90)
and
(event.event_type == 'credentials_update' and event.event_sub_type == 'password_reset')
and
event.challenge_requested != true
and
//Previous event
event.account_previous_activity_is_s == 1
and
//Account historical behavior
percentage_account_activity_time_of_day_3m <= max_account_activity_time_of_day_3m_th
and
event.count_succ_logins_per_accounttr >= min_account_activity_3m_th
```

Explanation

[DA-FR12] - Medium Risk - Login followed by a password reset at unusual hours
({event.event_occurred_at}).

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR13

Comment

Password change followed by new device request from an unusual location, different than customer location, in the last 24h

Actions

- [DA-FR13]

Variables

```
min_account_activity_1m =
digital_activity__rules_thresholds['min_account_activity_1m'].threshold; // min 10 logins per
week = 40 logins in 1 month

customer_region_to_lower_case =
com.feedzai.commons.udfs.StringTools.lowerCase(event.customer_region);
device_ip_region_to_lower_case =
com.feedzai.commons.udfs.StringTools.lowerCase(event.device_ip_region);
```

Condition

```
//Filters
((event.account_id ?? '' != '') and event.account_age ?? 0 > 30)
and
(event.event_type == 'device' and event.challenge_requested != true)
and
customer_region_to_lower_case != device_ip_region_to_lower_case
and
//Previous event
event.account_previous_change_is_pas == 1
and
//Account historical behavior
event.count_login_per_account_and_ip == 0
and
event.count_succ_logins_per_accounsr >= min_account_activity_1m
```

Explanation

[DA-FR13] - Medium Risk - Password change followed by new device request ({event.new_device_id}) from an unusual ({event.device_ip_region}) different than customer location ({event.customer_region}), in the last 24 hours

Mark As

The outcome of this rule is: **medium**